



Modelling Financial Markets

Seminar 4: Portfolio Optimisation — Hands-on



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Acronyms used in this material (1/2)

- ▣ AAPL: Apple Inc. (Nasdaq ticker)
- ▣ AI: Artificial Intelligence
- ▣ AMZN: Amazon.com, Inc. (Nasdaq ticker)
- ▣ BAC: Bank of America Corporation (NYSE ticker)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CSCO: Cisco Systems, Inc. (Nasdaq ticker)
- ▣ ERC: Equal Risk Contribution portfolio
- ▣ ETF: Exchange-Traded Fund
- ▣ GLS: Generalised Least Squares
- ▣ GME: GameStop Corp. (NYSE ticker)
- ▣ GMV: Global Minimum-Variance portfolio
- ▣ GMV-LW: Global Minimum-Variance portfolio with Ledoit–Wolf covariance
- ▣ GOOGL: Alphabet Inc., Class A shares (Nasdaq ticker)
- ▣ GS: The Goldman Sachs Group, Inc. (NYSE ticker)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HRP: Hierarchical Risk Parity
- ▣ HYG: iShares iBoxx High Yield Corporate Bond ETF
- ▣ IEF: iShares 7–10 Year Treasury Bond ETF
- ▣ JPM: JPMorgan Chase & Co. (NYSE ticker)
- ▣ LQD: iShares iBoxx Investment Grade Corporate Bond ETF
- ▣ ML: Machine Learning
- ▣ MS: Morgan Stanley (NYSE ticker)
- ▣ MSFT: Microsoft Corporation (Nasdaq ticker)

Acronyms used in this material (2/2)

- ▣ **MSTR: Strategy Inc., formerly MicroStrategy (Nasdaq ticker)**
- ▣ **MV: Mean-Variance tangency portfolio estimated from sample moments**
- ▣ **MV-LO: Mean-Variance tangency portfolio, Long-Only**
- ▣ **NVDA: NVIDIA Corporation (Nasdaq ticker)**
- ▣ **OLS: Ordinary Least Squares**
- ▣ **OOS: Out-Of-Sample**
- ▣ **US: United States**
- ▣ **WFC: Wells Fargo & Company (NYSE ticker)**

Outline

Introduction

Key Definitions

Part A: Derivations

Part B: Backtests and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive GMV, ERC and Black–Litterman portfolios and the bias of the in-sample Sharpe ratio
- ▣ Run an out-of-sample backtest with a rolling window, turnover and costs
- ▣ Test a difference in Sharpe ratios with HAC errors and a block bootstrap
- ▣ Judge whether an optimised portfolio really beats 1/N

- ▣ Background: Lecture 4; textbook: [Franke, Härdle & Hafner \(2019\)](#), Part I
- ▣ Seminar: Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Structure

- ▣ **Part A:** nine derivations (A1–A9)
- ▣ **Part B:** nine computational exercises, US and Bitcoin data (B1–B9)
- ▣ **Part C:** one open question on BVB portfolios, a starting point for the team project (C1), a critique of an AI answer (C2), and a replication of a published study on our data (C3)

How to Work in This Seminar

- ▣ **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- ▣ Every Part B exercise ends with an **interpretation question**
 - ▶ It cannot be answered by reading a number: ask what would change the conclusion
- ▣ Out of sample means: the weights of month t use only data up to month $t - 1$
- ▣ Report every Sharpe ratio difference with a p -value or an interval

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (GMV for N assets), A3 (ERC with two assets), A5 (Black–Litterman as GLS), A7 (variance of a Sharpe difference)
- Part B: B1 (1/ N versus Markowitz, sectors), B2 (shrinkage and GMV risk), B3 (ERC and HRP, sectors)

Proposed: you solve them

- Part A: A2 (model: A1); A4 (model: A3); A6 (model: A5); A8, A9 (model: A7)
- Part B: B4 (model: B2); B5 (model: B3); B6, B8 (model: B1); B7 (model: A5, A6); B9 (model: B1, A2)
- Part C: C1, open question (model: B1, B3); C2, critique of an AI answer (model: A7, B1); C3, replication of [Jensen, Kelly, Malamud & Pedersen \(2026\)](#) (model: B1, A7)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter4_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `matplotlib`
- Series: US sector and multi-asset ETFs, Bitcoin, eight BVB blue chips, BET-TR (course repository, `data/market`)
- One-month risk-free rate: [Kenneth French Data Library](#); USD/RON reference rate: [BNR](#)
- ETFs and stocks: dividend-adjusted prices; indices and Bitcoin: closing values
- Joint analyses: join the *prices* on common days first, then take month-end returns

Key Definitions: Portfolios

- Mean–variance (Markowitz (1952)): tangency $w_{MV} \propto \Sigma^{-1}\mu$; GMV $w_{GMV} \propto \Sigma^{-1}1$; both normalised to sum to one
- Two assets: $w^* = (\sigma_2^2 - \rho\sigma_1\sigma_2)/(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)$
- Risk contribution $RC_i = w_i(\Sigma w)_i/\sigma_p$; ERC (Maillard, Roncalli & Teïletche (2010)): equal RC_i
- HRP (López de Prado (2016)): correlation tree, then recursive bisection with inverse-variance weights
- Ledoit–Wolf (Ledoit & Wolf (2004)): $\hat{\Sigma} = \delta F + (1 - \delta)S$, constant-correlation target F
- Black–Litterman (Black & Litterman (1992)): prior $\pi = \delta_{RA}\Sigma w_b$, posterior μ_{BL} = precision-weighted mean of π and the views

Key Definitions: Evaluation and Inference

- ▣ Sharpe ratio $SR = \bar{r}^e / s(r^e)$, annualised $\times \sqrt{12}$; turnover $TO_t = \sum_i |w_{i,t} - w_{i,t-1}^+|$, $w^+ =$ drifted weights
- ▣ Net return: $r_t^{\text{net}} = r_t - c TO_t$, c the cost per unit of turnover as a decimal; a quote of 20 bp means $c = 0.002$ (1 bp = 0.01% = 10^{-4})
- ▣ i.i.d. test (Jobson & Korkie (1981); formula in Ledoit & Wolf (2008)), monthly SR_1, SR_2 , correlation ρ

$$\text{Var}(\widehat{SR}_1 - \widehat{SR}_2) \approx \frac{1}{T} \left[2 - 2\rho + \frac{1}{2} (SR_1^2 + SR_2^2 - 2SR_1SR_2\rho^2) \right]$$

- ▣ HAC test (Ledoit & Wolf (2008), Section 3.1): delta method with a Newey–West covariance of (r_1, r_2, r_1^2, r_2^2)
- ▣ Studentized circular block bootstrap (Ledoit & Wolf (2008), Section 3.2.2): pairs resampled in blocks, block length calibrated among $\{1, 2, 4, 6, 8, 10\}$, p -value from 4999 draws

Exercise A1: The GMV Portfolio for N Assets [Solved]

Task

- ▣ **Data and assumptions:** N risky assets with a known, positive definite covariance matrix Σ ; short sales allowed; weights sum to one. Numbers: SPY and TLT, monthly total returns August 2002 – July 2026 (data/market, $T = 288$), $\sigma_1 = 14.72\%$, $\sigma_2 = 13.49\%$, $\rho = -0.100$, means 11.73% and 4.32% a year
- ▣ (a) Write the Lagrangian of $\min_w w^\top \Sigma w$ subject to $w^\top \mathbf{1} = 1$ and derive $w_{\text{GMV}} = \Sigma^{-1} \mathbf{1} / A$, with $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$
- ▣ (b) Show that the minimum variance equals $\sigma_{\text{GMV}}^2 = 1/A$
- ▣ (c) Invert the 2×2 matrix and show that for $N = 2$ the weight of asset 1 is $w^* = (\sigma_2^2 - \rho\sigma_1\sigma_2) / (\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)$
- ▣ (d) Compute A , w^* , σ_{GMV} and the mean of the GMV mix for SPY and TLT
- ▣ **Report:** the derivation, the four numbers and one sentence on why the GMV volatility is below both assets

Exercise A1: Solution (1/2)

Solution: derivation

- (a) $\mathcal{L} = \mathbf{w}^\top \Sigma \mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{1} - 1)$; the first-order condition is $2\Sigma \mathbf{w} - \lambda \mathbf{1} = 0$, so $\mathbf{w} = \frac{\lambda}{2} \Sigma^{-1} \mathbf{1}$
- Imposing $\mathbf{1}^\top \mathbf{w} = 1$ gives $\frac{\lambda}{2} A = 1$, hence $\frac{\lambda}{2} = 1/A$ and $\mathbf{w}_{\text{GMV}} = \Sigma^{-1} \mathbf{1}/A$; the objective is strictly convex, so this is the unique minimum
- (b) $\sigma_{\text{GMV}}^2 = \mathbf{1}^\top \Sigma^{-1} \Sigma \Sigma^{-1} \mathbf{1}/A^2 = A/A^2 = 1/A$
- (c) With $\sigma_{12} = \rho\sigma_1\sigma_2$: $\Sigma^{-1} = \frac{1}{\det \Sigma} \begin{pmatrix} \sigma_2^2 & -\sigma_{12} \\ -\sigma_{12} & \sigma_1^2 \end{pmatrix}$, so $\Sigma^{-1} \mathbf{1} = \frac{1}{\det \Sigma} (\sigma_2^2 - \sigma_{12}, \sigma_1^2 - \sigma_{12})^\top$
- Dividing the first element by the sum of both gives $w^* = (\sigma_2^2 - \rho\sigma_1\sigma_2)/(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)$: the formula of the lecture recap is a special case

Exercise A1: Solution (2/2)

Solution: numbers

- ▣ Inputs: $\sigma_1^2 = 0.1472^2 = 0.02166$, $\sigma_2^2 = 0.1349^2 = 0.01820$, $\sigma_{12} = -0.100 \times 0.1472 \times 0.1349 = -0.00199$
- ▣ $\det \Sigma = 0.02166 \times 0.01820 - 0.00199^2 = 0.000390$; $\Sigma^{-1} \mathbf{1} = (0.02019, 0.02366)/0.000390 = (51.7, 60.6)$
- ▣ $A = 51.7 + 60.6 = 112.4$; $w^* = 51.7/112.4 = 0.461$ in SPY and 0.539 in TLT
- ▣ $\sigma_{\text{GMV}} = 1/\sqrt{112.4} = 9.43\%$; mean $0.461 \times 11.73\% + 0.539 \times 4.32\% = 7.73\%$
- ▣ Interpretation: with a negative correlation the two risks partly cancel, so the mix is 36% less volatile than SPY while keeping two-thirds of its mean

Exercise A2: Tangency Weights by Regression [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** X is the $T \times N$ matrix of excess returns, rows $r_t^\top$; $\hat{\mu} = X^\top 1/T$ and $\hat{\Sigma} = X^\top X/T - \hat{\mu}\hat{\mu}^\top$ (maximum likelihood); $\hat{\Sigma}$ is invertible
- ▣ (a) Write the OLS estimator $\hat{b}$ of the regression $1 = b^\top r_t + u_t$ (no constant term) as a function of $\hat{\mu}$ and $\hat{\Sigma}$
- ▣ (b) Apply Sherman–Morrison, $(A + uu^\top)^{-1} = A^{-1} - A^{-1}uu^\top A^{-1}/(1 + u^\top A^{-1}u)$, and show that $\hat{b} = \hat{\Sigma}^{-1}\hat{\mu}/(1 + \hat{\theta}^2)$, $\hat{\theta}^2 = \hat{\mu}^\top \hat{\Sigma}^{-1}\hat{\mu}$ (Britten-Jones (1999))
- ▣ (c) Show that $\hat{b}/1^\top \hat{b}$ equals the sample tangency weights when $1^\top \hat{\Sigma}^{-1}\hat{\mu} > 0$
- ▣ (d) State which hypothesis the OLS t -test of $b_i = 0$ tests, and write the restriction matrix R of the F -test of “tangency = $1/N$ ”
- ▣ **Report:** the proof in four lines and the two hypotheses; **Model:** A1

Exercise A3: ERC with Two Assets [Solved]

Task

- ▣ **Data and assumptions:** two assets, volatilities $\sigma_1, \sigma_2 > 0$, correlation $\rho \in (-1, 1)$, weights $w_1 + w_2 = 1$, long only; risk contribution $RC_i = w_i(\Sigma w)_i / \sigma_p$. Numbers: SPY ($\sigma_1 = 14.72\%$) and IEF ($\sigma_2 = 6.64\%$), $\rho = -0.100$, monthly August 2002 – July 2026
- ▣ (a) Write $RC_1\sigma_p$ and $RC_2\sigma_p$ explicitly for two assets
- ▣ (b) Prove that $RC_1 = RC_2$ if and only if $w_1\sigma_1 = w_2\sigma_2$, for any $\rho \in (-1, 1)$, so that ERC is inverse volatility
- ▣ (c) Compute the ERC weight of SPY and the volatility of the ERC portfolio; check that each asset carries half of the variance
- ▣ **Report:** the proof, w_1 , σ_{ERC} and the two risk shares

Exercise A3: Solution

Solution

- ▣ (a) $RC_1\sigma_p = w_1^2\sigma_1^2 + w_1w_2\rho\sigma_1\sigma_2$ and $RC_2\sigma_p = w_2^2\sigma_2^2 + w_1w_2\rho\sigma_1\sigma_2$
- ▣ (b) The cross terms are identical, so $RC_1 = RC_2 \iff w_1^2\sigma_1^2 = w_2^2\sigma_2^2 \iff w_1\sigma_1 = w_2\sigma_2$ (positive weights); ρ drops out
- ▣ (c) $1/\sigma_1 = 6.79$, $1/\sigma_2 = 15.06$: $w_1 = 6.79/(6.79 + 15.06) = 0.311$ in SPY, 0.689 in IEF; the numerical ERC solver gives the same 0.311
- ▣ $\sigma_p^2 = 0.311^2(0.02166) + 0.689^2(0.00441) + 2(0.311)(0.689)(-0.00098) = 0.00377$, $\sigma_{ERC} = 6.14\%$; risk shares 50% and 50%
- ▣ Interpretation: equal risk needs more than twice as much of the bond ETF, because it is less than half as volatile

Exercise A4: Who Carries the Risk in 60/40? [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** SPY and IEF as in A3, annualised $\sigma_1^2 = 0.02166$, $\sigma_2^2 = 0.00441$, $\sigma_{12} = -0.00098$; portfolio $w = (0.6, 0.4)$
- ▣ (a) Compute Σw , the volatility σ_p and the percentage risk shares $w_i(\Sigma w)_i / \sigma_p^2$ of the 60/40 portfolio
- ▣ (b) Compare σ_p with the ERC volatility of A3
- ▣ (c) Prove $\sum_i RC_i = \sigma_p$ with Euler's theorem for functions homogeneous of degree one
- ▣ (d) Prove that the ERC portfolio exists and is unique for Σ positive definite: show that $f(y) = \frac{1}{2}y^\top \Sigma y - \frac{1}{N} \sum_i \ln y_i$ is strictly convex on $y > 0$, tends to $+\infty$ at the boundary and at infinity, and that its first-order condition equalises the risk contributions
- ▣ **Report:** the numbers of (a)–(b) and the two proofs; **Model:** A3

Exercise A5: Black–Litterman as GLS [Solved]

Task

- **Data and assumptions:** prior $\pi = \mu + e_1$, views $q = P\mu + e_2$, $\text{Var}(e_1) = \tau\Sigma$, $\text{Var}(e_2) = \Omega$, e_1 and e_2 independent (Theil & Goldberger (1961)). Numbers: one asset, $\pi = 5\%$, $\sigma = 16\%$, $\tau = 0.05$, view $q = 8\%$ with standard deviation 2%
- (a) Stack the two equations as $y = Z\mu + e$ and derive the GLS estimator μ_{BL} and its covariance
- (b) Show that $P = I$ and $\Omega = \tau\Sigma$ give $\mu_{BL} = (\pi + q)/2$
- (c) For the one-asset numbers compute the two precisions, μ_{BL} , the weight of the view and the posterior standard deviation
- **Report:** the two formulas and the four numbers

Exercise A5: Solution

Solution

- (a) $y = (\pi^\top, q^\top)^\top$, $Z = (I, P^\top)^\top$, $V = \text{diag}(\tau\Sigma, \Omega)$; GLS: $\hat{\mu} = (Z^\top V^{-1}Z)^{-1}Z^\top V^{-1}y$
- Block multiplication: $\mu_{BL} = [(\tau\Sigma)^{-1} + P^\top \Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P^\top \Omega^{-1}q]$, covariance $[(\tau\Sigma)^{-1} + P^\top \Omega^{-1}P]^{-1}$: the Black-Litterman formula of the lecture
- (b) $P = I$, $\Omega = \tau\Sigma$: $[2(\tau\Sigma)^{-1}]^{-1}(\tau\Sigma)^{-1}(\pi + q) = (\pi + q)/2$: equal precisions, equal weights
- (c) Prior variance $\tau\sigma^2 = 0.05 \times 0.0256 = 0.00128$, precision 781; view variance $0.02^2 = 0.0004$, precision 2500
- $\mu_{BL} = (781 \times 5\% + 2500 \times 8\%)/3281 = 7.29\%$; view weight $2500/3281 = 0.76$; posterior standard deviation $1/\sqrt{3281} = 1.75\%$
- Interpretation: the view is three times as precise as the prior, so it gets three-quarters of the weight; the posterior is more precise than either source

Exercise A6: The Sector View of the Lecture [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** nine sector ETFs, monthly excess returns August 2016 – July 2026, Σ annualised; benchmark $1/N$, $\delta_{RA} = 2.5$, $\tau = 0.05$; implied XLK minus XLU = $P\pi = 2.22\%$, $P\Sigma P^\top = 0.0477$
- ▣ (a) Write the prior distribution of the view portfolio $P\mu$ implied by $\mu \sim N(\pi, \tau\Sigma)$
- ▣ (b) With the view XLK minus XLU = 3% and $\Omega = \tau P\Sigma P^\top$ (He & Litterman (2002)), show that the posterior view is the simple average of prior and view, and compute it
- ▣ (c) Compute the new weights $w_{BL} = (\delta_{RA}\Sigma)^{-1}\mu_{BL}$ and say which weights change
- ▣ **Report:** the posterior view and the XLK and XLU weights; **Model:** A5

Exercise A7: The Variance of a Sharpe Difference [Solved]

Task

- ▣ **Data and assumptions:** two return series, i.i.d. bivariate Normal, means μ_i , variances σ_i^2 , correlation ρ , $SR_i = \mu_i/\sigma_i$ monthly. Numbers: sector backtest, $T = 271$ months (January 2004 – July 2026), monthly Sharpe GMV 0.2051, $1/N$ 0.1908, $\rho = 0.694$
- ▣ (a) Write the asymptotic covariance matrix of $\sqrt{T}(\hat{\mu}_1, \hat{\mu}_2, \hat{\sigma}_1^2, \hat{\sigma}_2^2)$ under Normality
- ▣ (b) Apply the delta method to $f = \mu_1/\sigma_1 - \mu_2/\sigma_2$ and derive $T \text{Var}(\widehat{SR}_1 - \widehat{SR}_2) \rightarrow 2 - 2\rho + \frac{1}{2}(SR_1^2 + SR_2^2 - 2SR_1SR_2\rho^2)$
- ▣ (c) Test $H_0: SR_{\text{GMV}} = SR_{1/N}$ at $\alpha = 5\%$ (two-sided z-test) and compare with the HAC p -value 0.75
- ▣ **Report:** the variance formula, z , p , the decision and its meaning

Exercise A7: Solution (1/2)

Solution: derivation

- (a) Normal: the asymptotic covariance of $\sqrt{T}(\hat{\mu}_1 - \mu_1, \hat{\mu}_2 - \mu_2, \hat{\sigma}_1^2 - \sigma_1^2, \hat{\sigma}_2^2 - \sigma_2^2)$ has entries σ_i^2 and σ_{12} (means), $2\sigma_i^4$ and $2\sigma_{12}^2$ (variances), and zero between means and variances
- (b) Gradient of μ_i/σ_i : $\partial/\partial\mu_i = 1/\sigma_i$, $\partial/\partial\sigma_i^2 = -SR_i/(2\sigma_i^2)$
- Mean block: $\sigma_1^2/\sigma_1^2 + \sigma_2^2/\sigma_2^2 - 2\sigma_{12}/(\sigma_1\sigma_2) = 2 - 2\rho$
- Variance block: $\frac{SR_1^2}{4\sigma_1^4}2\sigma_1^4 + \frac{SR_2^2}{4\sigma_2^4}2\sigma_2^4 - 2\frac{SR_1SR_2}{4\sigma_1^2\sigma_2^2}2\sigma_{12}^2 = \frac{1}{2}(SR_1^2 + SR_2^2 - 2SR_1SR_2\rho^2)$; add the two blocks

Exercise A7: Solution (2/2)

Solution: test

- ▣ (c) Bracket: $2 - 2(0.694) + 0.5(0.2051^2 + 0.1908^2 - 2 \times 0.2051 \times 0.1908 \times 0.694^2) = 0.612 + 0.5(0.0421 + 0.0364 - 0.0377) = 0.6324$
- ▣ Standard error $\sqrt{0.6324/271} = 0.0483$; statistic $z = (0.2051 - 0.1908)/0.0483 = 0.30$; $p = 2[1 - \Phi(0.30)] = 0.77$
- ▣ Decision: $p = 0.77 > 0.05$, do not reject H_0 ; the HAC test agrees ($p = 0.75$)
- ▣ Interpretation: the annualised gap, 0.71 against 0.66, is well inside sampling noise; 22 years of data cannot rank GMV and 1/N

Exercise A8: ERC versus 1/N [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** the same sector backtest as A7, $T = 271$; monthly Sharpe ERC 0.2018, 1/N 0.1908, correlation 0.9927; i.i.d. Normal returns
- ▣ (a) Compute the bracket of A7, the standard error, z and the two-sided p -value; decide at $\alpha = 5\%$
- ▣ (b) Show that the bracket tends to $\frac{1}{2}(SR_1 - SR_2)^2$ as $\rho \rightarrow 1$
- ▣ (c) Compare with A7: explain why a smaller difference is here closer to significance
- ▣ **Report:** z , p , the decision, the limit and a two-sentence explanation; **Model:** A7

Exercise A9: The In-Sample Sharpe Bias [Proposed]

Solution discussed in class

Task

- **Data and assumptions:** i.i.d. $N(\mu, \Sigma)$ returns, maximum-likelihood $\hat{\mu}$, $\hat{\Sigma}$; known facts: $\hat{\mu} \sim N(\mu, \Sigma/T)$, $\mathbb{E}[\hat{\Sigma}^{-1}] = \frac{T}{T-N-2}\Sigma^{-1}$ for $T > N + 2$, $\hat{\mu}$ independent of $\hat{\Sigma}$. Numbers: nine sectors, $T = 60$, true $\theta = 0.66$ a year
- (a) Compute $\mathbb{E}[\hat{\theta}^2 \mid \hat{\mu}]$ for $\hat{\theta}^2 = \hat{\mu}^\top \hat{\Sigma}^{-1} \hat{\mu}$
- (b) Compute $\mathbb{E}[\hat{\mu}^\top \Sigma^{-1} \hat{\mu}]$ with the trace formula and conclude $\mathbb{E}[\hat{\theta}^2] = (T\theta^2 + N)/(T - N - 2)$ (Kan & Zhou (2007))
- (c) Evaluate it for $N = 9$, $T = 60$, $\theta = 0.66/\sqrt{12}$ monthly; convert to an annual Sharpe ratio and compare with the simulated median in-sample Sharpe of 1.56 (lecture)
- (d) Give the unbiased estimator of θ^2 implied by the formula
- **Report:** the derivation, the number and the corrected estimator; **Model:** A7

Exercise B1: 1/N versus Markowitz on Sectors [Solved]

Task

- **Data and assumptions:** nine US sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY; data/market, adjusted closes), month-end simple returns January 1999 – July 2026, minus the one-month Treasury bill ([Kenneth French Data Library](#))
- (a) Each month t estimate 1/N, MV (tangency), MV-LO and GMV on months $t - 60, \dots, t - 1$ only; out of sample January 2004 – July 2026 ($T = 271$)
- (b) Compute the annualised Sharpe ratio (monthly $\times \sqrt{12}$) and the mean monthly turnover $\sum_i |w_{i,t} - w_{i,t-1}^+|$ of each rule
- (c) Test $H_0: SR_{\text{rule}} = SR_{1/N}$ with the HAC test ([Ledoit & Wolf \(2008\)](#), Section 3.1; Bartlett kernel, 4 lags) and the studentized circular block bootstrap (Section 3.2.2; block from $\{1, 2, 4, 6, 8, 10\}$ by Algorithm 3.1, 4999 draws), $\alpha = 5\%$
- **Report:** a table with Sharpe, turnover, $\hat{\Delta}$ and the two p -values; the bootstrap interval for MV-LO

Interpretation question

- MV-LO beats 1/N by 0.06 with $p = 0.60$: how long a sample would make this difference significant?

Exercise B1: Solution (1/2)

Solution: procedure and results

- Procedure: rolling 60-month estimation, one-month holding, drifted weights w^+ for turnover; MV and GMV from sample moments, MV-LO by quadratic programming with $0 \leq w_i \leq 1$

	Sharpe	Turnover	$\hat{\Delta}$	HAC p	Bootstrap p (block)
1/N	0.66	0.02	—	—	—
MV	0.26	3.86	−0.40	0.22	0.30 (6)
MV-LO	0.72	0.23	+0.06	0.60	0.63 (1)
GMV	0.71	0.23	+0.05	0.75	0.76 (1)

- Reading: MV has a volatility of 69% and lost all wealth once; its turnover of 3.86 a month means trading almost four times the portfolio
- The three other rules land within 0.06 of 1/N; no p -value is below 0.05: do not reject H_0 for any rule

Exercise B1: Solution (2/2)

Solution: interpretation

- MV-LO: $\hat{\Delta} = 0.063$, HAC standard error 0.120; HAC $p = 0.60$; studentized bootstrap (block 1) $p = 0.63$, 95% interval $[-0.19, 0.31]$ covers zero: do not reject
- Significance at 5% needs a standard error of $0.063/1.96 = 0.032$; standard errors fall like $1/\sqrt{T}$
- Required length: $22.6 \times (0.120/0.032)^2 \approx 22.6 \times 14 \approx 310$ years of out-of-sample data
- Conclusion: the estimated gain of MV-LO is small, but its interval admits both losses and gains up to 0.31; failure to reject does not establish equality or economic equivalence

Exercise B2: Does Shrinkage Lower GMV Risk? [Solved]

Task

- ▣ **Data and assumptions:** the backtest of B1 (nine sectors, 60-month window, OOS January 2004 – July 2026); GMV with the sample covariance versus GMV-LW, the Ledoit–Wolf estimator with the constant-correlation target (Ledoit & Wolf (2004))
- ▣ (a) Compute the out-of-sample volatilities of GMV, GMV-LW and $1/N$
- ▣ (b) Compute the variance ratio $\text{Var}(\text{GMV-LW}) / \text{Var}(\text{GMV})$ and its 95% percentile interval from a circular block bootstrap (Künsch (1989); 6-month blocks, 5000 draws)
- ▣ (c) Compare the mean monthly turnover of the two rules
- ▣ (d) On the last window (August 2021 – July 2026) count the correlation eigenvalues outside the Marchenko–Pastur band $[(1 - \sqrt{c})^2, (1 + \sqrt{c})^2]$, $c = N/T$
- ▣ **Report:** volatilities, ratio with interval, turnover, the eigenvalue count

Interpretation question

- ▣ Why is the gain small with 9 assets and 60 months? When would it be larger?

Exercise B2: Solution

Solution

- ▣ (a) Volatility: GMV 11.90%, GMV-LW 11.80%, 1/N 14.28%: both GMV rules cut risk by about a sixth
- ▣ (b) Variance ratio $0.1180^2/0.1190^2 = 0.984$; bootstrap interval $[0.918, 1.062]$ contains 1: no significant reduction at 5%
- ▣ (c) Turnover falls from 0.234 to 0.137 (-41%): the real benefit of shrinkage here is stability, not risk
- ▣ (d) $c = 9/60 = 0.15$, band $[0.38, 1.92]$; eigenvalues 5.25, 1.35, 0.84, 0.50, 0.32, $\dots$, 0.11: one above (the market), five below
- ▣ Interpretation: with $N/T = 0.15$ the sample matrix is already decent; the gain from shrinkage grows with N/T and with a target close to the truth

Exercise B3: ERC and HRP on Sectors [Solved]

Task

- ▣ **Data and assumptions:** nine sector ETFs; in (a)–(b) weights from the last 60-month window only (August 2021 – July 2026, sample covariance), to compare the allocations
- ▣ (a) Compute the ERC weights (convex problem of [Maillard, Roncalli & Teïletche \(2010\)](#)) and the HRP weights ([López de Prado \(2016\)](#): single-linkage tree on the Euclidean distances between the columns of $D = (\sqrt{(1 - \rho_{ij})/2})$, recursive bisection)
- ▣ (b) Compute the percentage risk shares of $1/N$ on the same window
- ▣ (c) Run the rolling backtest of B1 (60-month window, monthly) for ERC and HRP; test their Sharpe ratios against $1/N$ with the HAC test at $\alpha = 5\%$; report turnover and the correlation with $1/N$
- ▣ **Report:** the largest and smallest weights, the risk shares, $\hat{\Delta}$, ρ and turnover

Interpretation question

- ▣ ERC and $1/N$ have correlation 0.993: is ERC a different strategy or a small tilt?

Exercise B3: Solution

Solution

- (a) ERC: XLP 14.8%, XLV 13.9%, XLU 12.5% ... XLB 8.6%; HRP: XLP 18.4%, XLU 17.1%, XLV 15.2% ... XLB 4.5%
- (b) 1/N risk shares: XLB 13.8%, XLI 13.4% ... XLP 7.8%: equal weights, unequal risks; ERC shifts weight to the defensive sectors
- (c) ERC: Sharpe 0.70 vs 0.66, $\hat{\Delta} = +0.04$, $p = 0.09$; HRP: 0.70, $\hat{\Delta} = +0.04$, $p = 0.32$: neither rejects H_0 at 5%
- Turnover: 1/N 0.024, ERC 0.028, HRP 0.082; correlation with 1/N: ERC 0.993, HRP 0.979
- Interpretation: ERC is a small, cheap tilt towards low-volatility sectors; HRP is a larger tilt that trades three times as much

Exercise B4: Shrinkage on 16 ETFs [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** combined universe, nine sectors plus EFA, EEM, TLT, IEF, LQD, HYG, GLD (data/market); common months May 2007 – July 2026; 60-month window, OOS May 2012 – July 2026 ($T = 171$)
- ▣ (a)–(c) Repeat B2 (a)–(c): volatilities, variance ratio with the same bootstrap, turnover
- ▣ (d) Report the mean shrinkage intensity δ and the condition numbers of the sample and shrunk matrices on the last window
- ▣ (e) Estimate Σ by analytical nonlinear shrinkage (Ledoit & Wolf (2020)), build the resulting GMV portfolio and compare its out-of-sample return variance with GMV and GMV-LW, with bootstrap intervals
- ▣ **Model:** B2

Interpretation question

- ▣ Is constant correlation a sensible target for stocks and Treasuries together?

Exercise B5: Risk Parity with Bitcoin [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** the eight multi-asset ETFs (SPY, EFA, EEM, TLT, IEF, LQD, HYG, GLD) with and without Bitcoin (BTC-USD.CC, closing values), prices joined on common days, monthly from August 2010; 60-month window, OOS August 2015 – July 2026 ($T = 132$)
- ▣ (a) Compute the full-sample volatility of Bitcoin and its risk share in the 1/N portfolio of nine assets
- ▣ (b) Backtest 1/N, ERC and HRP with and without Bitcoin; report Sharpe ratios and volatilities, and test each with-minus-without Sharpe difference with the HAC test at $\alpha = 5\%$
- ▣ (c) Report the mean Bitcoin weight in ERC and HRP
- ▣ **Model:** B3; literature: [Petukhina, Trimborn, Härdle & Elendner \(2021\)](#), [Pele et al. \(2023\)](#)

Interpretation question

- ▣ Would you expect the past Bitcoin premium to persist? What about survivorship?

Exercise B6: HRP versus GMV on 16 ETFs [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** the combined universe and backtest of B4 (OOS May 2012 – July 2026, $T = 171$)
- ▣ (a) Compare HRP and GMV: Sharpe ratio, volatility, turnover and net Sharpe at 50 bp per unit of turnover
- ▣ (b) Test $H_0: SR_{\text{HRP}} = SR_{\text{GMV}}$ with the HAC test and with the studentized block bootstrap of B1, at $\alpha = 5\%$ and 10%
- ▣ **Report:** the four statistics for both rules, $\hat{\Delta}$, both p -values and the bootstrap interval; **Model:** B1

Interpretation question

- ▣ GMV wins on volatility, HRP on Sharpe: which criterion fits which investor?

Exercise B7: Black–Litterman on a Multi-Asset Set [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** the eight multi-asset ETFs, monthly excess returns August 2016 – July 2026 ($T = 120$), Σ annualised; benchmark $1/N$, $\delta_{RA} = 2.5$, $\tau = 0.05$
- ▣ (a) Compute the implied returns $\pi = \delta_{RA}\Sigma w_b$ and the implied GLD minus TLT
- ▣ (b) Add the view GLD minus TLT = 4% a year with Ω as in [He & Litterman \(2002\)](#); compute the posterior view and the new weights
- ▣ (c) Compare with the sample mean of GLD minus TLT
- ▣ **Model:** A5, A6

Interpretation question

- ▣ The weights move far although the view is modest: why?

Exercise B8: The Break-Even Cost [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** combined universe of B4, MV-LO versus $1/N$; net return $r_t - c TO_t$ with c the decimal cost per unit of turnover, quoted in basis points ($c = \text{bp} \times 10^{-4}$)
- ▣ (a) Compute the gross Sharpe ratios, the HAC p -value of their difference and the mean turnovers
- ▣ (b) Find the value c^* at which the net Sharpe ratios are equal, on a grid of 0–200 bp with step 0.1 bp
- ▣ (c) Check c^* with the approximation $c^* \approx \Delta SR / [12(TO_1/\sigma_1 - TO_2/\sigma_2)]$ (annual σ ; decimal c^* , times 10^4 in bp; assumes costs leave the volatilities almost unchanged)
- ▣ **Model:** B1

Interpretation question

- ▣ Retail costs on BVB or US ETFs: is MV-LO worth implementing?

Exercise B9: Is the Tangency Portfolio 1/N? [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** nine sectors, monthly excess returns January 1999 – July 2026 ($T = 331$); Britten-Jones regression of 1 on r_t , no constant term (A2)
- ▣ (a) Test $H_0: b_1 = \dots = b_9$ (tangency = 1/N) with the exact $F(8, T - 9)$ test at $\alpha = 5\%$
- ▣ (b) Repeat with a HAC Wald test, χ^2_8 , Bartlett kernel with $\lfloor 4(T/100)^{2/9} \rfloor$ lags
- ▣ (c) Repeat (a)–(b) on the halves January 1999 – September 2012 and October 2012 – July 2026
- ▣ **Report:** a 3×2 table of statistics and p -values; **Model:** B1, A2

Interpretation question

- ▣ When the two tests disagree, which one do you trust with $T = 166$ and 8 restrictions?

C1: An Optimal Portfolio of BVB Blue Chips in RON [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** eight BVB blue chips (TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL) and BET-TR from data/market, October 2014 – August 2026; foreign ETFs (SPY, EFA, IEF, GLD) converted to RON at the [BNR](#) USD/RON reference rate on common days; monthly returns, 36-month window
- ▣ (a) Build 1/N, GMV-LW, ERC and HRP portfolios of the eight blue chips, then of the blue chips plus the four foreign ETFs, rebalanced monthly out of sample
- ▣ (b) Compare each portfolio with BET-TR (never BET): Sharpe ratio, volatility, maximum drawdown
- ▣ (c) Test each global rule against BET-TR and against its BVB-only version (HAC test, $\alpha = 5\%$), with a Holm adjustment over one family of eight tests (four rules $\times$ two benchmarks)
- ▣ (d) Repeat (b) with a RON risk-free rate, or show how the Sharpe ratios change with it
- ▣ **Report:** a Quantlet folder in the team repository and one slide with the conclusion and its main limitation

The answer should also discuss

- ▣ Currency risk of the foreign legs, transaction costs and taxes for a Romanian investor

C2: Critique an AI Answer [Proposed]

Solution discussed in class

The prompt sent to an artificial intelligence (AI) assistant

- “prices holds daily adjusted closes of ten sector ETFs. Write Python code for a GMV backtest with monthly rebalancing and give me its annualised Sharpe ratio.”

The assistant's answer

- “Below is an out-of-sample GMV backtest, rebalanced monthly to the GMV weights:”

```
import numpy as np
r = prices.resample('ME').last().pct_change().dropna()
S = r.cov().values
w = np.linalg.solve(S, np.ones(len(S)))
w = w / w.sum()
port = r.dot(w)
sr = port.mean() / port.std() * np.sqrt(252)
print(f"Annualised Sharpe ratio: {sr:.2f}")
```

monthly simple returns
covariance matrix
$\Sigma^{-1} \mathbf{1}$
GMV weights, sum to one
portfolio return

C2: Critique an AI Answer, the Task [Proposed]

Solution discussed in class

- The code runs without an error message, but it contains **three** mistakes typical of backtests
 - ▶ Everything else in the answer is correct
- For each mistake
 - ▶ (a) Name the line and explain in one or two sentences why it is wrong
 - ▶ (b) Say how you would detect it: a test, a number from A7 or B1, or a slide of the lecture
 - ▶ (c) Write the corrected line(s) of Python
- Log each mistake as in `AI_ERRORS.md`
 - ▶ What was asked, what the AI answered, what was wrong, how you found out, the correct version
- Hint: reread the slide “Key Definitions: Evaluation and Inference” and the rule on out-of-sample weights

C3: The Implementable Efficient Frontier on Our Data [Proposed]

Solution discussed in class

Context, data and assumptions

- Jensen, Kelly, Malamud & Pedersen (2026): learning the weights directly, with trading costs in the objective, gives the highest net utility among ML portfolio rules (Lecture 4, case study)
- The shrinkage rules of Part B (Ledoit & Wolf (2017)) ignore that costs grow with trade size; does the ranking of Table 2 hold on a small universe?
- 14 US stocks and 9 sector ETFs from data/market (JPM, BAC, C, GS, MS, WFC, NVDA, AAPL, MSFT, AMZN, GOOGL, GME, CSCO, MSTR and XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY), prices joined on common days; monthly excess returns over the one-month rate of [Kenneth French Data Library](#)
- Five signals at each month end, ranked into $[0, 1]$ across assets (Section 5.1.2): returns of the last month and of months $t - 6$ to $t - 1$ and $t - 12$ to $t - 1$, 1-year volatility of daily returns, log of the 6-month average daily dollar volume
- Costs (eq. 35): $\Lambda_i = 0.2/V_i$, V_i the 6-month average daily dollar volume; wealth \$10 billion at the end of 2020, growing with SPY; $\gamma = 10$

Interpretation question

- If Portfolio-ML earns a lower net Sharpe ratio than $1/N$ but a higher utility, which of the two numbers answers the question of an investor with $\gamma = 10$, and for which γ would the ranking flip?

C3: The Implementable Efficient Frontier, the Task [Proposed]

Solution discussed in class

- (a) Estimate Σ_t each month from the daily returns of the past 10 years with exponential weights, half-lives of 378 days for correlations and 126 days for variances (Section 5.1.5), and compute m_t from eq. (14) by fixed-point iteration
- (b) Build Portfolio-ML as in Proposition 4 and eq. (40): random Fourier features on the grid of Table 1, the signals of the last 12 months in eq. (24), hyperparameters chosen each year by the realised utility of validation backtests since January 2011, out of sample from January 2014
- (c) Build the rivals on the same months: 1/N, minimum variance, Markowitz-ML (eq. 32, returns forecast by a ridge regression on the same random features) and Static-ML (eq. 34 with $\phi = 1$)
- (d) Report for every rule the columns of Table 2 (return, volatility, gross and net Sharpe ratios, cost, utility, turnover, leverage) and the posterior probabilities of Table 3
- (e) Extension: repeat Portfolio-ML for a wealth of \$1, 10 and 100 billion at the end of 2020 and $\gamma \in \{1, 5, 10, 20, 100\}$, and plot the implementable frontiers as in Figure 1(B), with 1/N as a point
- **Report:** two tables in the layout of Tables 2 and 3, the frontier figure, and a Quantlet folder in the team repository
- **Model:** B1 (out-of-sample backtest), A7 (test of a difference in performance)

Common Pitfalls

Design

- ▣ Estimating weights with data from the evaluation month (look-ahead)
- ▣ Reporting in-sample Sharpe ratios of optimised portfolios
- ▣ Ignoring turnover and costs
- ▣ Comparing with BET instead of BET-TR

Statistics

- ▣ Comparing Sharpe ratios without a test
- ▣ i.i.d. tests on dependent returns
- ▣ Many rules and universes, one p -value below 0.05
- ▣ Inverting a near-singular covariance matrix

- ▣ Rule: state the window, the rebalancing frequency, the costs and the benchmark of every backtest

Take-Home Assignment (Team Project)

- Choose **ten** assets: at least three BVB stocks, two bond ETFs and one commodity or crypto asset
 - ▶ One reporting currency (RON, as in C1: foreign prices converted at the BNR rate on common days) and a risk-free rate in that currency
- Backtest 1/N, GMV-LW, ERC and HRP with a rolling window
 - ▶ Report Sharpe ratio, volatility, maximum drawdown and turnover, gross and net of 20 bp
 - ▶ Test each rule against 1/N (HAC and block bootstrap)
- Sensitivity: window of 36 versus 60 months, compared on the same out-of-sample months
- Log every AI error you catch in AI_ERRORS.md (at least three per project)
- Deliverables: a GitHub repository with a notebook that runs top-to-bottom and a 2-page summary

Further Study on Quantinar

Advanced courses

- ▣  Hierarchical Clustering
- ▣  Minimum Spanning Tree

Code

- ▣ **Notebook:** `chapter4_seminar_notebook.ipynb` (Colab)
- ▣ **Quantlets:** github.com/danpele/MFM/tree/main/Quantlets/Ch_04

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